Activity 4

If line l_1 is perpendicular to line l_2 and line l_3 has slope 3 then

(i) find the equation of line l_1

(ii) find the equation of line l_2

(iii) find the equation of line l_3

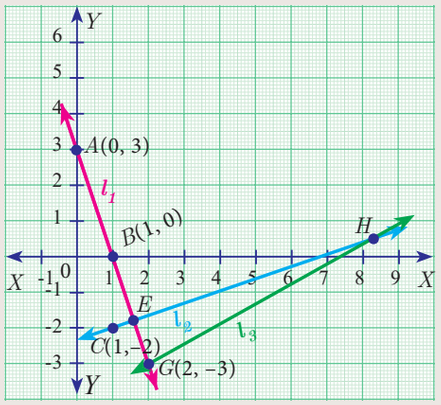
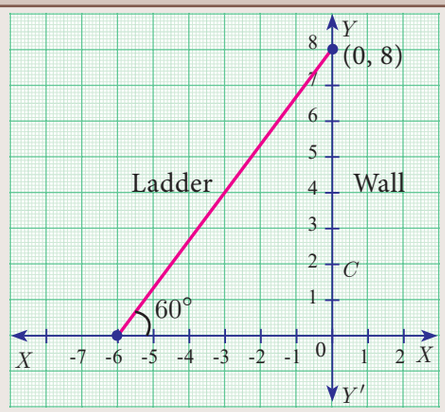


Fig. 5.38

Activity 5

A ladder is placed against a vertical wall with its foot touching the horizontal floor. Find the equation of the ladder under the following conditions.

No.	Condition	Picture	Equation of the ladder
(i)	The ladder is inclined at 60° to the floor and it touches the wall at (0,8)	 Fig. 5.39	_____
(ii)	The foot and top of the ladder are at the points (2,4) and (5,1)	_____	_____

Exercise 5.3

1. Find the equation of a straight line passing through the mid-point of a line segment joining the points $(1,-5)$, $(4,2)$ and parallel to (i) X axis (ii) Y axis
2. The equation of a straight line is $2(x-y)+5=0$. Find its slope, inclination and intercept on the Y axis.
3. Find the equation of a line whose inclination is 30° and making an intercept -3 on the Y axis.
4. Find the slope and y intercept of $\sqrt{3}x+(1-\sqrt{3})y=3$.
5. Find the value of 'a', if the line through (–2,3) and (8,5) is perpendicular to $y=ax+2$

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